

# Gaurav Gupta

SCIENTIST/ENGINEER-SC

Space Application Centre, Ahmedabad

☎ (+91) 9083722796 | ✉ gauravg.aero@gmail.com | 🏠 <https://gauravity30.github.io/> | 📺 gauravity30 | 🌐 gauravgupta030 | 📧 Gaurav Gupta

## Summary

Aerospace engineer at the Space Applications Centre (ISRO) specializing in computational fluid dynamics and biological fluid mechanics. Primary experience encompasses investigating avian flight aerodynamics, modeling microgravity two-phase flows, and developing custom flow-solver subroutines in Fortran and Python, alongside engineering precision optomechanical systems for opto-quantum communication payloads. Key research interests focus on fluid mechanics of animal locomotion, biological flows, fluid-structure interactions, and low-gravity fluid mechanics.

## Engineering Research Experience

### Space Application Centre, Indian Space Research Organization (ISRO)

Ahmedabad, Gujarat

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Jul. 2025 - Present

#### Opto-Mechanical Engineering for Opto-Quantum Payloads

- Responsible for engineering space-qualified optomechanical subsystems for quantum communication payloads.
- Led precision assembly, integration, and sub-micron alignment protocols for sensitive quantum optics setups.
- Developed and implemented automated workflows and CAD/CAE pipelines for opto-mechanical design and STOP analysis using Siemens NX Open APIs.

#### Microgravity Fluid Mechanics

- Formulated the conceptual system design and architecture for a space-based recirculating aquatic life support payload.
- Conducted two-phase simulation to investigate the surface tension driven fluid flow and transport in microgravity environment using OpenFOAM.
- Investigating stability of an air-water interface under microgravity using OpenFOAM.

### Indian Institute of Technology, Kharagpur

Kharagpur, India

RESEARCH INTERN

Mar. 2023 - May. 2025

- Explored various high-fidelity computational methods to investigate the aerodynamics of avian tails in gliding flight.
- High-fidelity DNS (for 2D) and hybrid RANS-LES methods such as DES were identified as suitable computational methods for investigating the aerodynamics of avian flight.
- Conducted high-fidelity DNS (using Incompact3D and NEK5000), DES, and RANS (using SU2) of a representative common swift geometry at Re 10000 - 20000.
- Pursued under the guidance of Dr. Sandeep Saha, IIT KGP and Dr. Manu K V, IIST.

## Education

### Indian Institute of Space Science and Technology

Thiruvananthapuram, India

B.TECH. IN AEROSPACE ENGINEERING

Dec. 2021 - Jun. 2025

- Cumulative Grade Point Average (CGPA): 8.28/10
- Relevant coursework: Aerodynamics, Computational Fluid Dynamics, Aircraft Design, Boundary Layer Theory, Finite Element Method, Flight Dynamics & Control, Introduction to Astrobiology
- Relevant Labs: Aerodynamics Lab, Modelling and Simulation Lab, Flight Mechanics Lab, and Programming

### Kendriya Vidyalaya No.1 AFS Kalaikunda

Kharagpur, India

HIGH SCHOOL (XI & XII)

Jul. 2019 - May. 2021

- Graduated as the topper of school with 97.4% in the CBSE AISSCE examination.

## Open-Source Contributions

### Incompact3D

CFD, Fortran

HIGH-ORDER FINITE DIFFERENCE FLOW SOLVER

GitHub

- Developed a fortran subroutine to calculate lift and drag forces for a 3D immersed body simulations using Incompact3D.
- Working on improving the performance of importing STL files for high-fidelity simulations using Incompact3D.

### PyMech

CFD, Python, NEK5000

A PYTHON SOFTWARE SUITE FOR NEK5000 AND SIMSON

GitHub

- Improving the in-built open\_dataset function to support unstructured datasets produced using NEK5000.
- Used for post-processing of high-fidelity simulation data using NEK5000.

## TurbCourse

CFD, Python, NEK5000

A COURSE IN TURBULENCE SIMULATION

GitHub

- Improved the performance of POD and DMD modules for high-fidelity simulation data using NEK5000.
- Added capability for conducting frequency analysis of the POD modes.
- Uses PyMech for reading/writing of NEK5000 data.

## FreeFEM-Euler

CFD, FreeFEM++, GMSH

FREEFEM BASED INCOMPRESSIBLE EULER SOLVER.

Jun. 2024

- An incompressible inviscid flow solver based on Artificial Compressibility Method was developed using FreeFEM++.
- The code is capable of solving 2D problems based on in-built FreeFEM++ and GMSH meshes.
- Open-source and available on GitHub.

## Skills

### Modelling and Computer Aided Design

DS Solidworks, DS Catia, Autodesk Fusion 360, Autodesk Autocad, Blender, Siemens NX

### Flow Solvers and Mesh Generation

Incompact3D, NEK5000, SU2, OpenFOAM, GMSH, Coreform Cubit, Pointwise

### Programming and Numerical Libraries

C++, Fortran, FreeFEM++, Julia, Python, MATLAB

### HPC and Tools

Linux, Git, LaTeX, Visit, Paraview

## Publications

### Enabling automation in Opto-Quantum Communication Program through APIs

Internal Proceedings/Non-Public

GUPTA, GAURAV, SHARMA, ANKIT

2026

ISRO/DOS Seminar on Computer Technology, Cyber Security and Mission IT 2026

## Certificates

Ongoing **Fundamentals of Fluid-Solid Interactions**, École Polytechnique and Coursera

Apr. 2024 **High Performance Computing**, IIT Bombay and NPTEL

Dec, 2022 **Machine Learning with Python (Honors)**, IBM and Coursera

Oct, 2022 **Design of Fixed Wing Unmanned Aerial Vehicle (Topper)**, IIT Kanpur and NPTEL

Nov, 2022 **Aircraft Design**, IIT Bombay and NPTEL

## Extracurricular

- Directed IIST's technical festival (Conscientia) for 1000+ participants as Chief Coordinator, managing a INR 2.2 million budget and securing an INR 400,000 profit.
- Led team initiatives in designing fixed-wing and rotor aircraft as Student Coordinator, while orchestrating UAV design workshops and competitions.